

Biomaterials

生物材料

南方科技大学
材料科学与工程系
2024年春季课程

2.9 SURFACE MODIFICATION OF MATERIALS USED IN MEDICINE

Much effort goes into the design, synthesis, and fabrication of biomaterials and devices to ensure that they have the appropriate mechanical properties, durability, and functionality. To cite a few examples, a hip joint should withstand high stresses, a hemodialyzer should have the requisite permeability characteristics, and the pumping bladder in an artificial heart should flex for millions of cycles without failure. The bulk structure of the materials governs these properties.



Unmodified surface



Overcoat

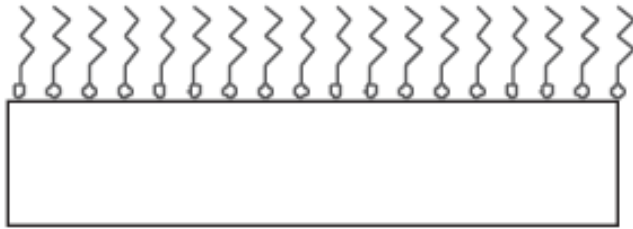
- Solvent coat
- Grafted or adsorbed surface layer
- Metallization
- Sprayed hydroxyapatite (flame or electrostatic)



Surface gradient

- Graft
- Interpenetrating network
- Ion implant

2.9 SURFACE MODIFICATION OF MATERIALS USED IN MEDICINE

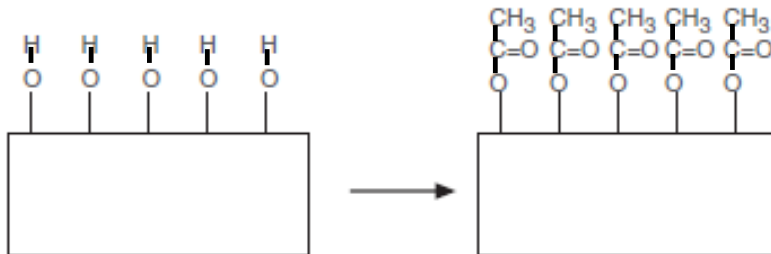


Self assembled film, Langmuir-Blodgett overlayer

- N-Alkyl thiols on gold
- N-Alky silanes on silica
- N-Alky phosphates on Ti
- Multilayers are possible



Surface active bulk additive



Surface chemical reaction

- Oxidation
- Fluorination
- Silanization



Etching and roughening

Surface chemical reaction is also frequently observed



Polyelectrolyte multilayer films

TABLE 1 Some Physicochemically Surface-Modified Biomaterials

To modify blood compatibility

Octadecyl group attachment to surfaces (albumin affinity)

Silicone-containing block copolymer additive

Plasma fluoropolymer deposition

Plasma siloxane polymer deposition

Radiation grafted hydrogel

Chemically modified polystyrene for heparin-like activity

To influence cell adhesion and growth

Oxidized polystyrene surface

Ammonia plasma-treated surface

Plasma-deposited acetone or methanol film

Plasma fluoropolymer deposition (reduce endothelial adhesion to IOLs)

Intraocular Lens (contact lens)

To control protein adsorption

Surface with immobilized poly(ethylene glycol) (reduce adsorption)

Treated ELISA dish surface (increase adsorption)

Affinity chromatography column

Surface cross-linked contact lens (reduce adsorption)

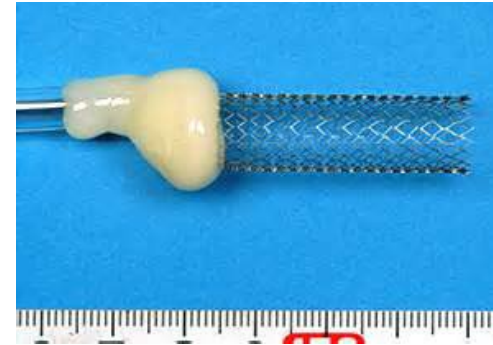
TABLE 1 Some Physicochemically Surface-Modified Biomaterials

To improve lubricity

Plasma treatment

Radiation grafting (hydrogels)

Interpenetrating polymeric networks



To improve wear resistance and corrosion resistance

Ion implantation

Diamond deposition

Anodization



To alter transport properties

Polyelectrolyte grafting

To modify electrical characteristics

Polyelectrolyte grafting

Magnetron sputtering of titanium

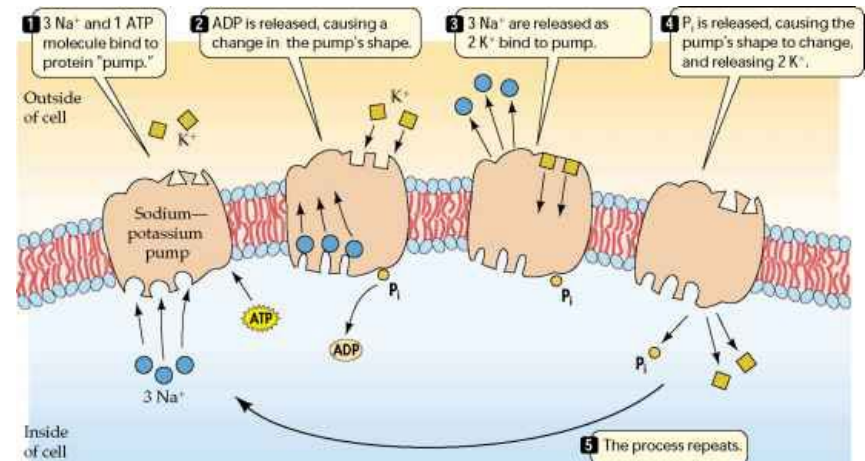


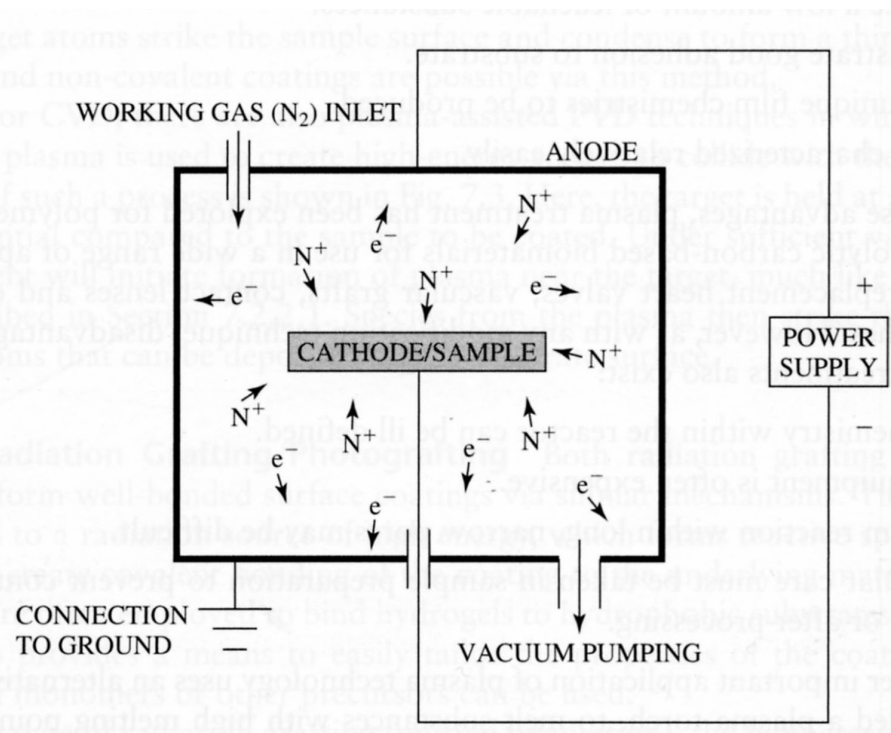
TABLE 2 Physical and Chemical Surface Modification Methods

	Polymer	Metal	Ceramic	Glass
Noncovalent coatings				
Solvent coating	✓	✓	✓	✓
Langmuir–Blodgett film deposition	✓	✓	✓	✓
Surface-active additives	✓	✓	✓	✓
Vapor deposition of carbons and metals ^a	✓	✓	✓	✓
Vapor deposition of parylene (<i>p</i> -xylylene)	✓	✓	✓	✓
Covalently attached coatings				
Radiation grafting (electron accelerator and gamma)	✓	—	—	—
Photografting (UV and visible sources)	✓	—	—	✓
Plasma (gas discharge) (RF, microwave, acoustic)	✓	✓	✓	✓
Gas-phase deposition				
• Ion beam sputtering	✓	✓	✓	✓
• Chemical vapor deposition (CVD)	—	✓	✓	✓
• Flame spray deposition	—	✓	✓	✓
Chemical grafting (e.g., ozonation + grafting)	✓	✓	✓	✓
Silanization	✓	✓	✓	✓
Biological modification (biomolecule immobilization)	✓	✓	✓	✓
Modifications of the original surface				
Ion beam etching (e.g., argon, xenon)	✓	✓	✓	✓
Ion beam implantation (e.g., nitrogen)	—	✓	✓	✓
Plasma etching (e.g., nitrogen, argon, oxygen, water vapor)	✓	✓	✓	✓
Corona discharge (in air)	✓	✓	✓	✓
Ion exchange	✓ ^b	✓	✓	✓
UV irradiation	✓	✓	✓	✓
Chemical reaction				
• Nonspecific oxidation (e.g., ozone)	✓	✓	✓	✓
• Functional group modifications (oxidation, reduction)	✓	—	—	—
• Addition reactions (e.g., acetylation, chlorination)	✓	—	—	—
Conversion coatings (phosphating, anodization)	—	✓	—	—
Mechanical roughening and polishing	✓	✓	✓	✓

TABLE 3 Biomedical Applications of Glow Discharge
Plasma-Induced Surface Modification Processes

- A. Plasma treatment (etching)
 - 1. Clean
 - 2. Sterilize
 - 3. Cross-link surface molecules
 - B. Plasma treatment (etching) and plasma deposition
 - 1. Form barrier films
 - a. Protective coating
 - b. Electrically insulating coating
 - c. Reduce absorption of material from the environment
 - d. Inhibit release of leachables
 - e. Control drug delivery rate
 - 2. Modify cell and protein reactions
 - a. Improve biocompatibility
 - b. Promote selective protein adsorption
 - c. Enhance cell adhesion
 - d. Improve cell growth
 - e. Form nonfouling surfaces
 - f. Increase lubricity
 - 3. Provide reactive sites
 - a. For grafting or polymerizing polymers
 - b. For immobilizing biomolecules
-

Methods of surface coating: Plasma Discharge



Charged particles are attracted to the sample surface, which acts as the cathode.

Particles may be positive or negative ions, free radicals, electrons, atoms, molecules or photons.

Often used to add OH or NH₂ groups to surface as a precursor to further modification

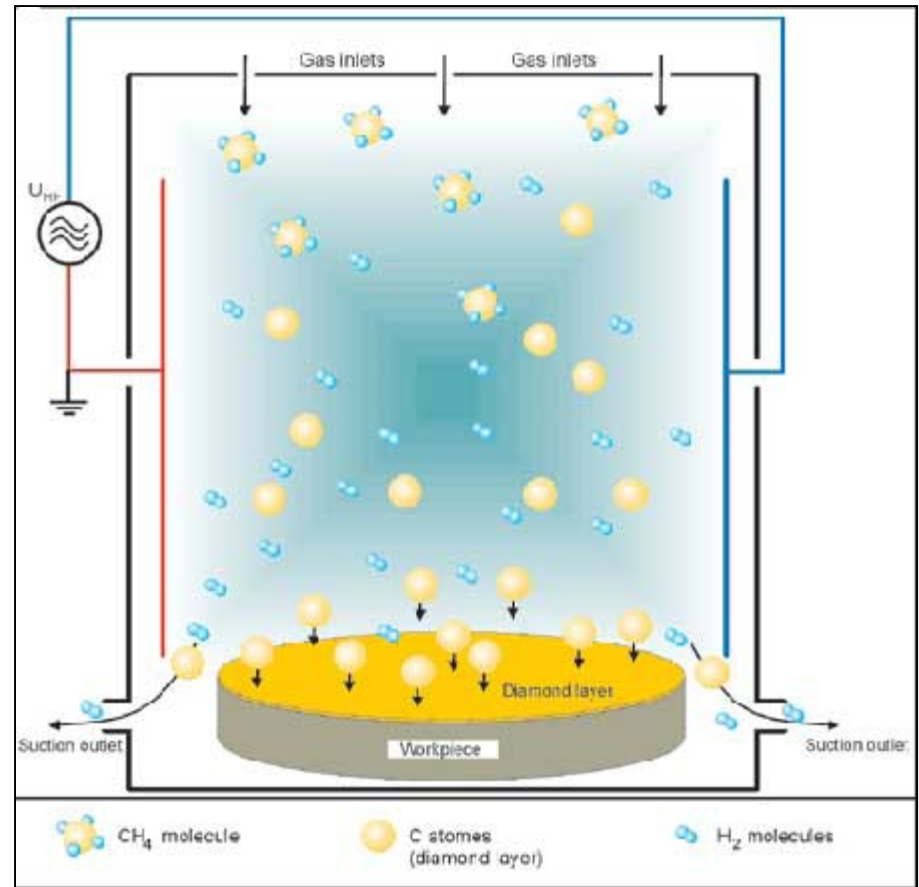
Plasma Discharge

- Advantages:
 - Coatings are conformal
 - Free of voids/pinhole defects
 - Easily prepared
 - Sterile when removed from reactor
 - Produce low amount of leachable substances
 - Demonstrate good adhesion to substrate
 - Allow unique film chemistries to be produced
 - Easily characterized
- Disadvantages
 - Chemistry within reactor may be undefined
 - Equipment often expensive
 - Uniform reaction within long, narrow pores may be difficult
 - Care must be taken in sample preparation to prevent contamination

Vapor Deposition: Chemical (CVD)

In Chemical Vapor Deposition (CVD) a reactive gas is passed over the substrate to be coated, inside of a heated, environmentally controlled reaction chamber.

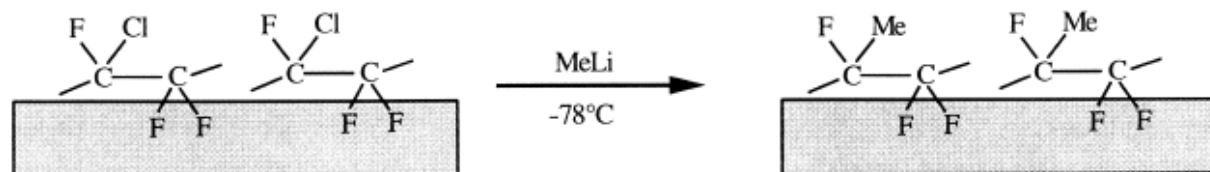
In this case (right) CH_4 gas is introduced to create a diamond-like coating



Some specific surface chemical modifications

A

Alkylation of poly(chlorotrifluoroethylene)



B

Trifluoroacetic anhydride reaction of a hydroxylated surface



C

Glycidyl group introduction into a polysiloxane

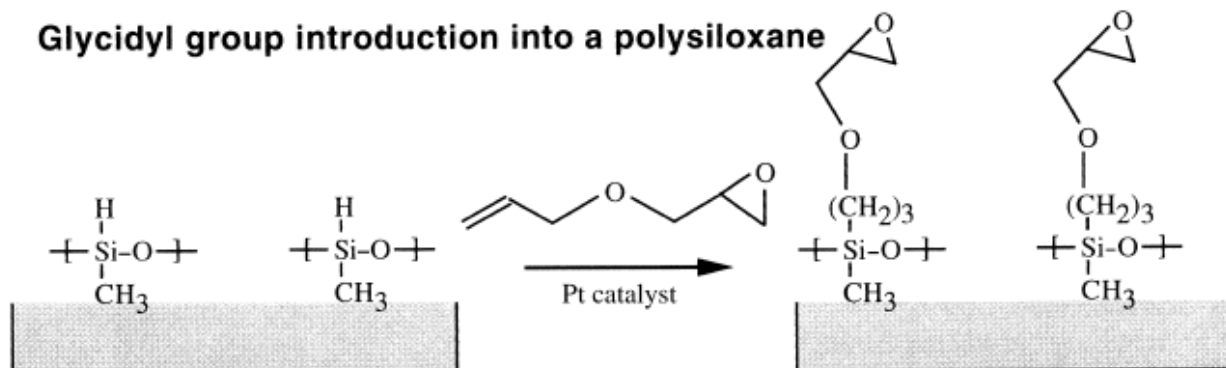


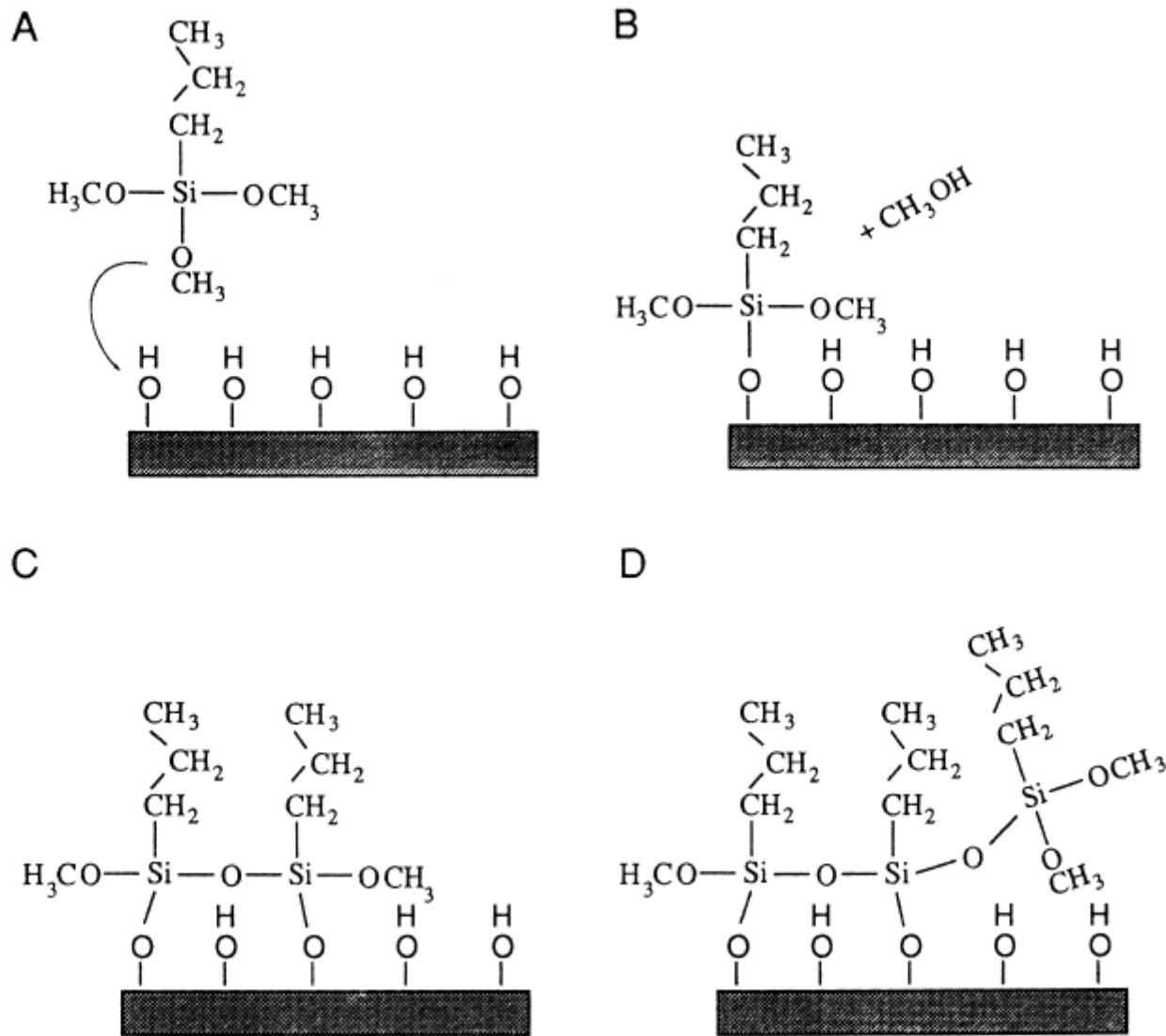
FIG. 3. Some specific chemical reactions to modify surfaces.

Silanes for surface modification of biomaterials

TABLE 4 Silanes for Surface Modification of Biomaterials

$ \begin{array}{c} \text{X} \\ \\ \text{X} - \text{Si} - \text{R} \\ \\ \text{X} \end{array} $	
X = leaving group	R = functional group
-Cl	-(CH ₂) _n CH ₃
-OCH ₃	-(CH ₂) ₃ NH ₂
-OCH ₂ CH ₃	-(CH ₂) ₂ (CF ₂) ₅ CF ₃
	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{-(CH}_2\text{)}_3\text{O-C-C=CH}_2 \\ \\ \text{O} \end{array} $
	-CH ₂ CH ₂ - 

The chemistry of a typical silane surface modification reaction

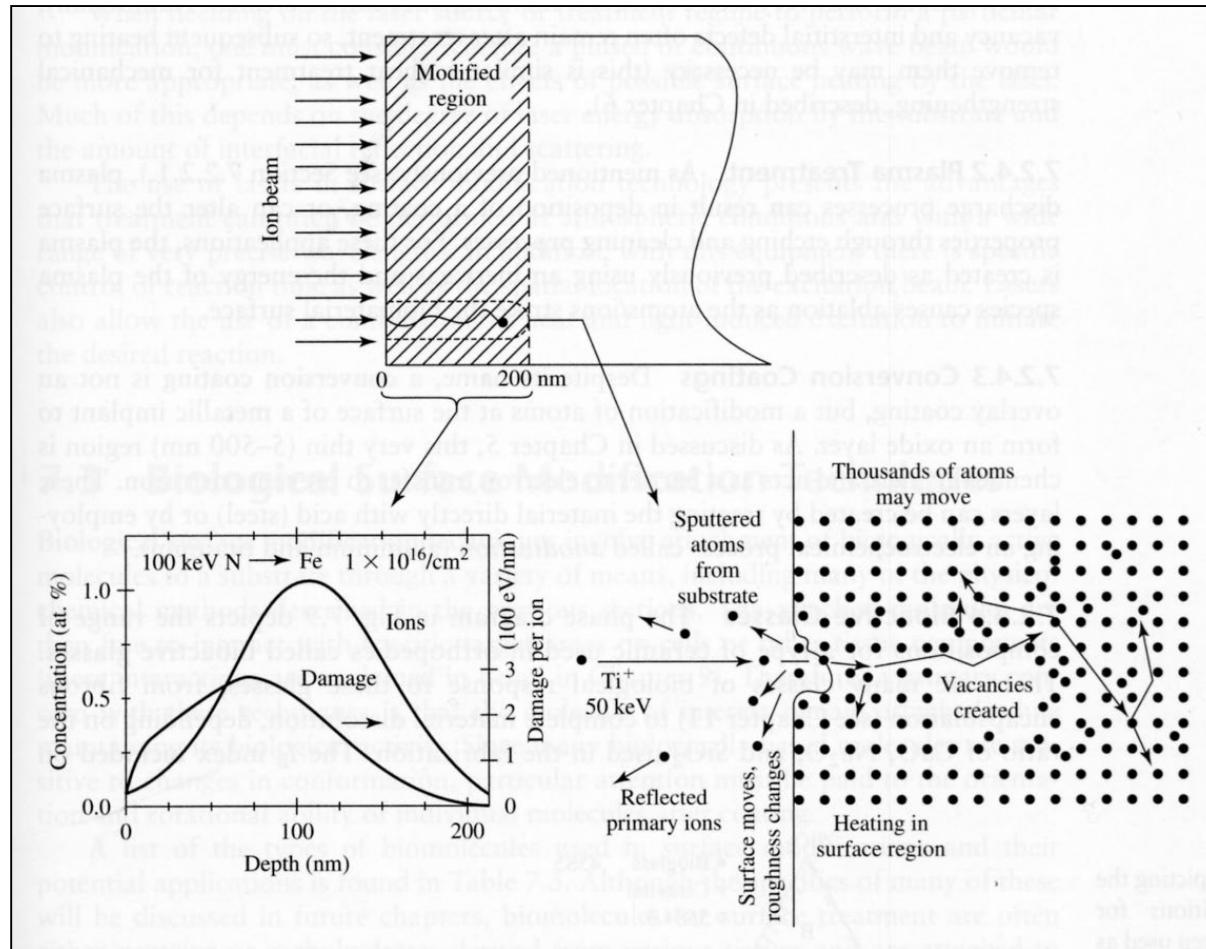


The chemistry of a typical silane surface modification reaction.

Radiation Grafting and Photografting

- Substrate is exposed to a radiation source of high energy, which forms a reactive species at the surface to create covalent bonding of the coating to the underlying material
- Often employed to bind hydrogels to hydrophobic substrates
- Biomaterial substrate may be placed in a monomer solution then irradiated by electrons or gamma rays to form a polymerized coating.

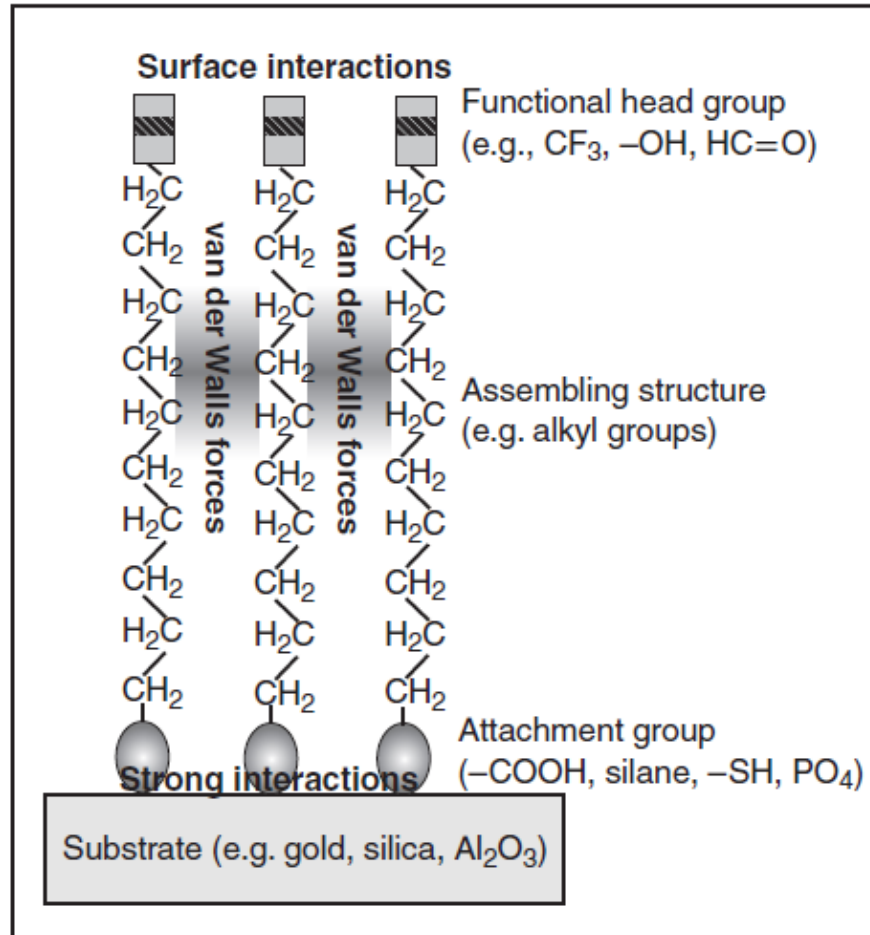
Ion beam implantation



This method can create surfaces with high hardness, wear, corrosion resistance and biocompatibility

It can also cause surface damage in the form of sputtering of surface atoms, surface roughness and changes in the crystal structure.

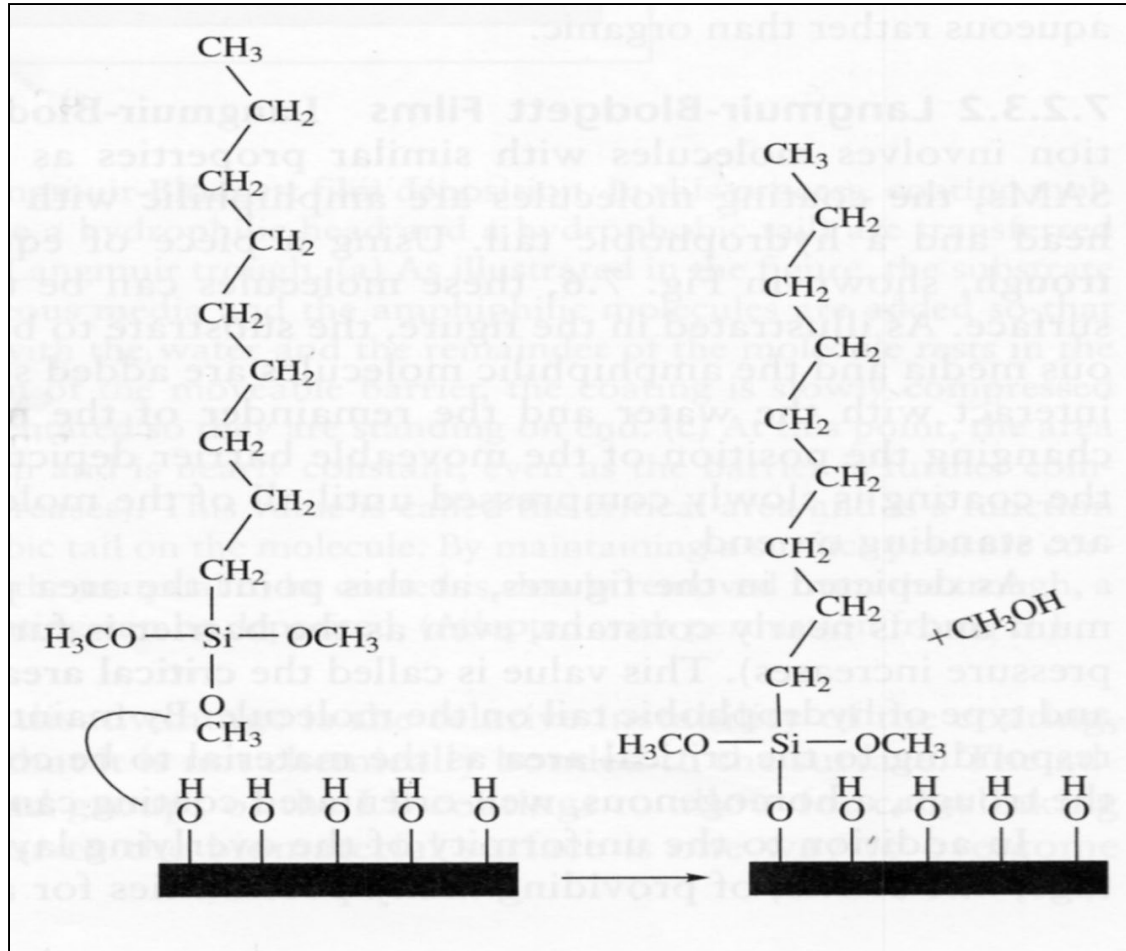
Self-Assembled Monolayers (SAMs)



SAMs are **amphiphilic**, having both hydrophilic (polar) and hydrophobic (nonpolar) parts. They are made up of 3 parts:

- The attachment group
- A long hydrocarbon chain
- A functional (polar) head group

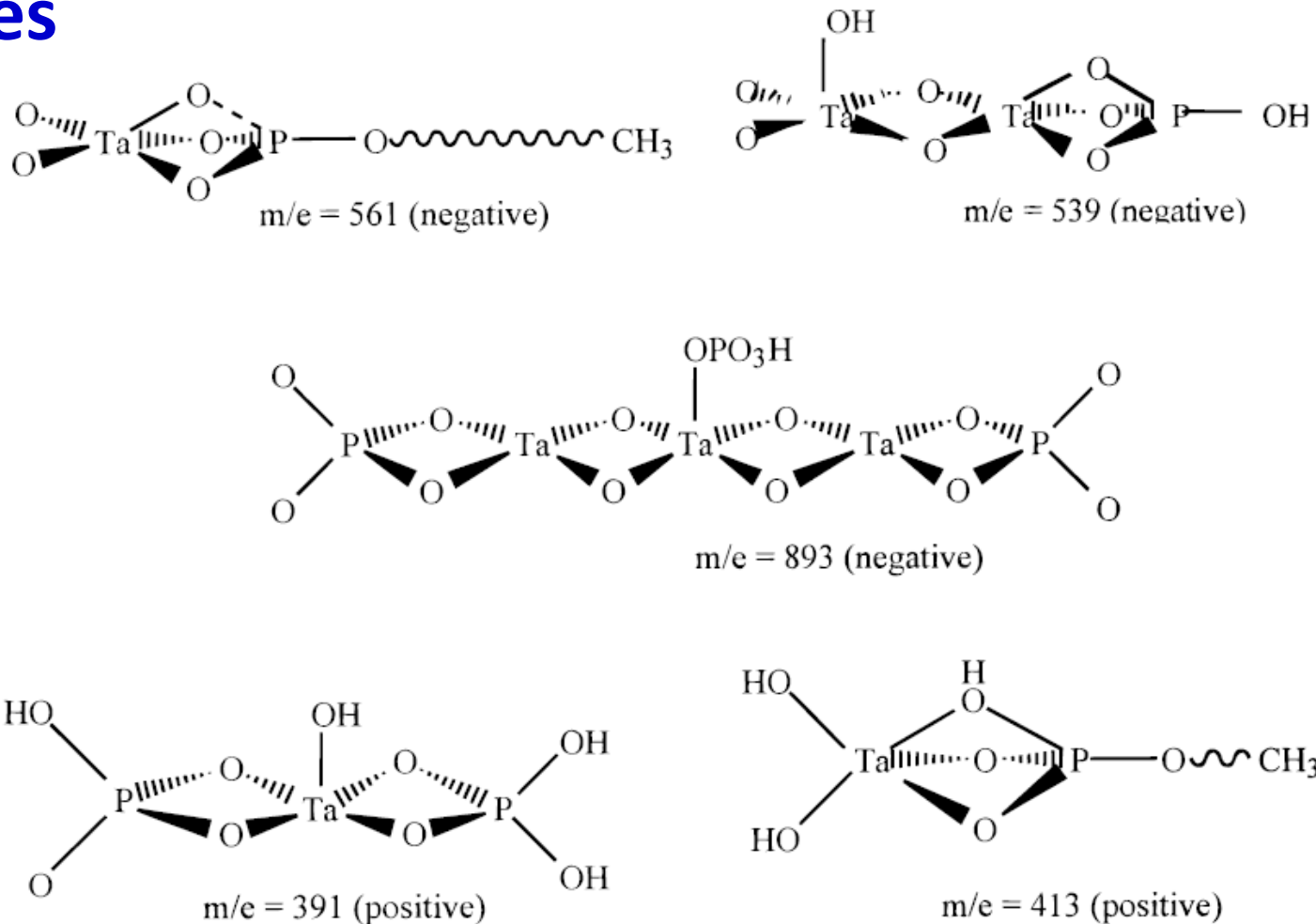
Self-Assembled Monolayers (SAMs)



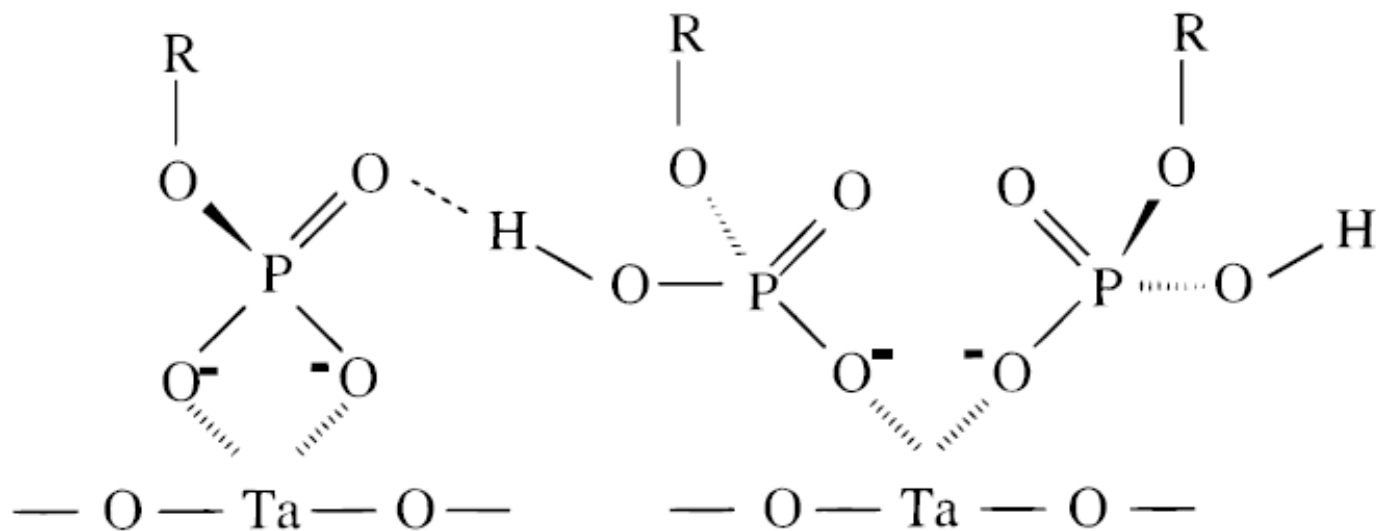
In the picture, hydroxyl groups form a strong attachment to the substrate.

A strong exothermic reaction attaches the Silane to the OH

Structural Chemistry of Self-Assembled Monolayers of Octadecylphosphoric Acid on Tantalum Oxide Surfaces

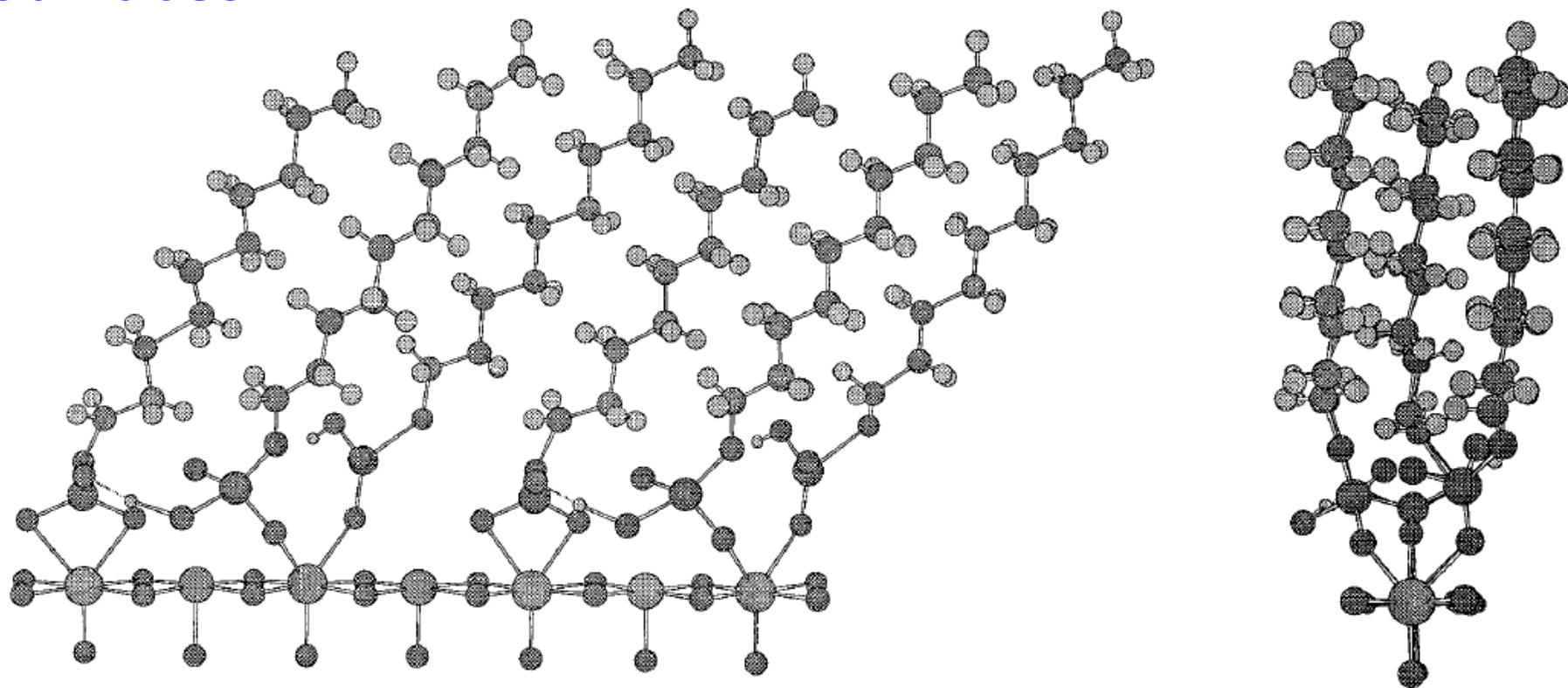


Proposed structures for some of the observed fragments in the negative and positive ToF-SIMS spectra of the ODP SAM monolayer on tantalum oxide (Ta₂O₅).



Bidentate (type A, left) and monodentate (type B, right) phosphate coordination to tantalum ions, with the possibility for the formation of intermolecular hydrogen bonding.

Structural Chemistry of Self-Assembled Monolayers of Octadecylphosphoric Acid on Tantalum Oxide Surfaces



Ball-and-stick model of six adjacent ODP molecules with monodentate and bidentate coordination (in the ratio of 2:1) to the Ta₂O₅ substrate surface (see Figure 8 for binding of the phosphate group to the substrate). Only 10 carbon atoms of the alkane chain are shown for the sake of simplicity. The tilt angle of the hydrocarbon chain is approximately 30° relative to the surface normal.

Bioinspired catecholic chemistry for surface modification



Mefp5

1 SSEEYKGGYYPGNAYHYHSGGSYHG
 26 SGYHGGYKGKYYGKAKKYYYKYKNS
 51 GKYYLKKARKYHRKGYYKYYGGSS

Mefp3

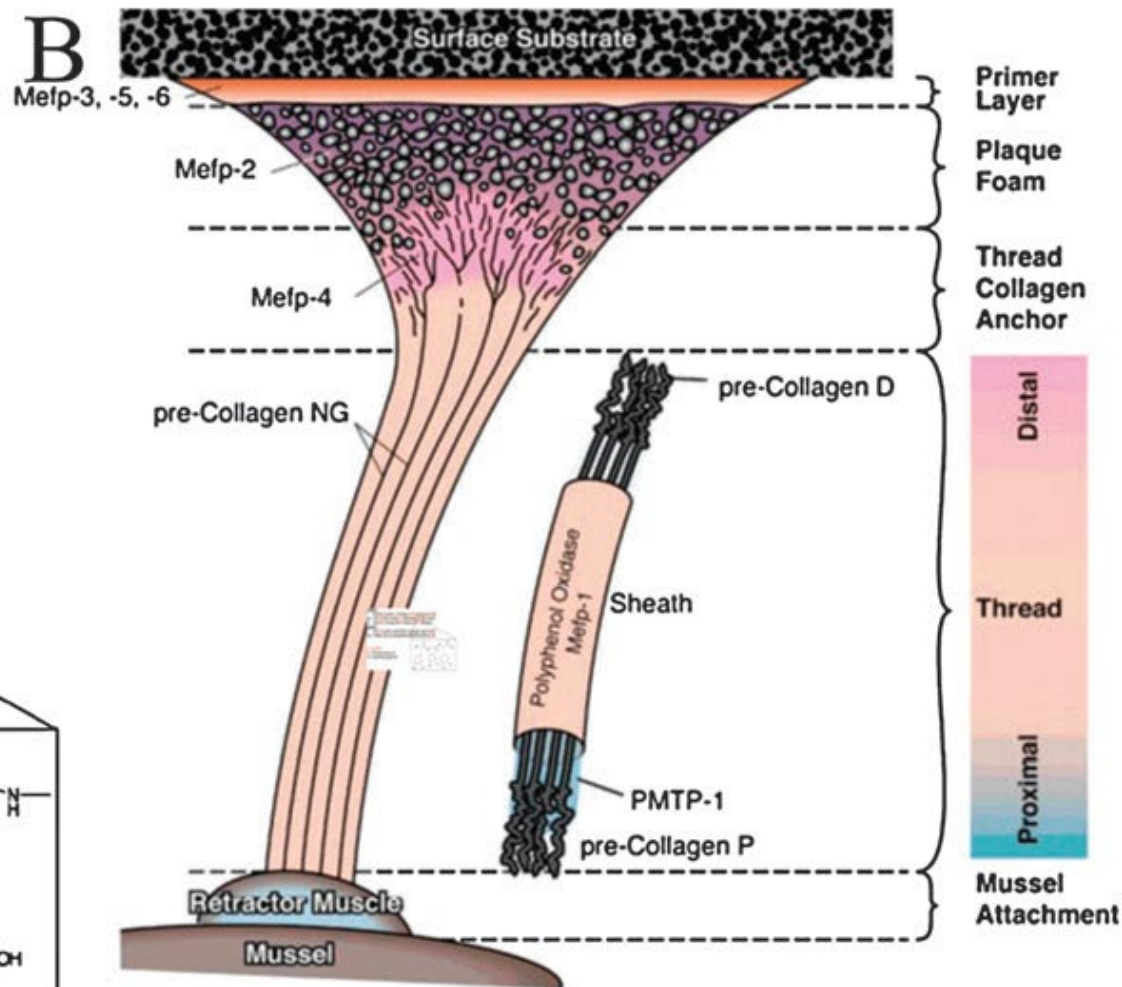
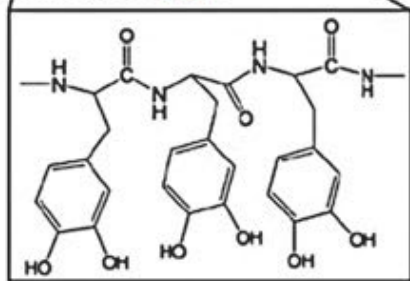
1 ADYYGPNYGPPRRYGGGNYNRYNRY
 26 GRRYGGYKGNNGWNRGRRGKYW

Y: DOPA

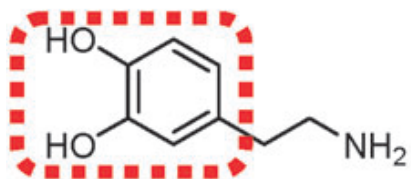
S: Phosphoserine

R: Hydroxyarginine

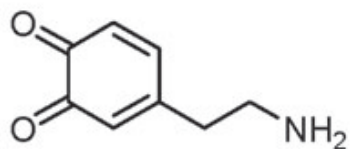
C



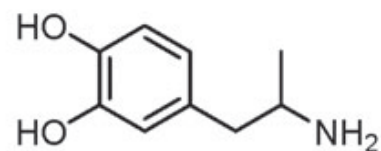
Dopamine and its analogues



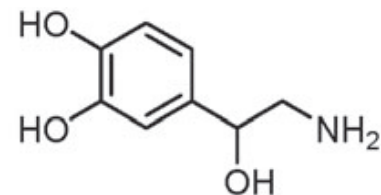
dopamine



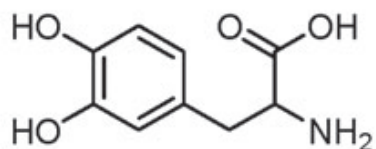
dopamine-quinone



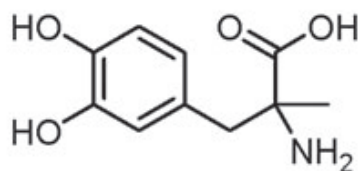
alpha-methyldopamine



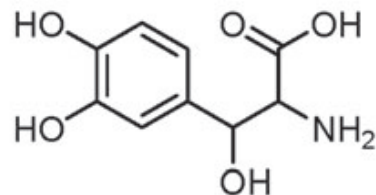
norepinephrine



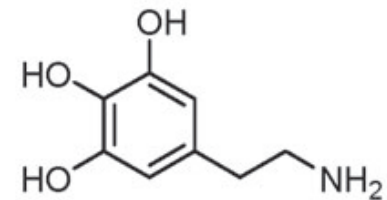
DOPA



alpha-methyldopa

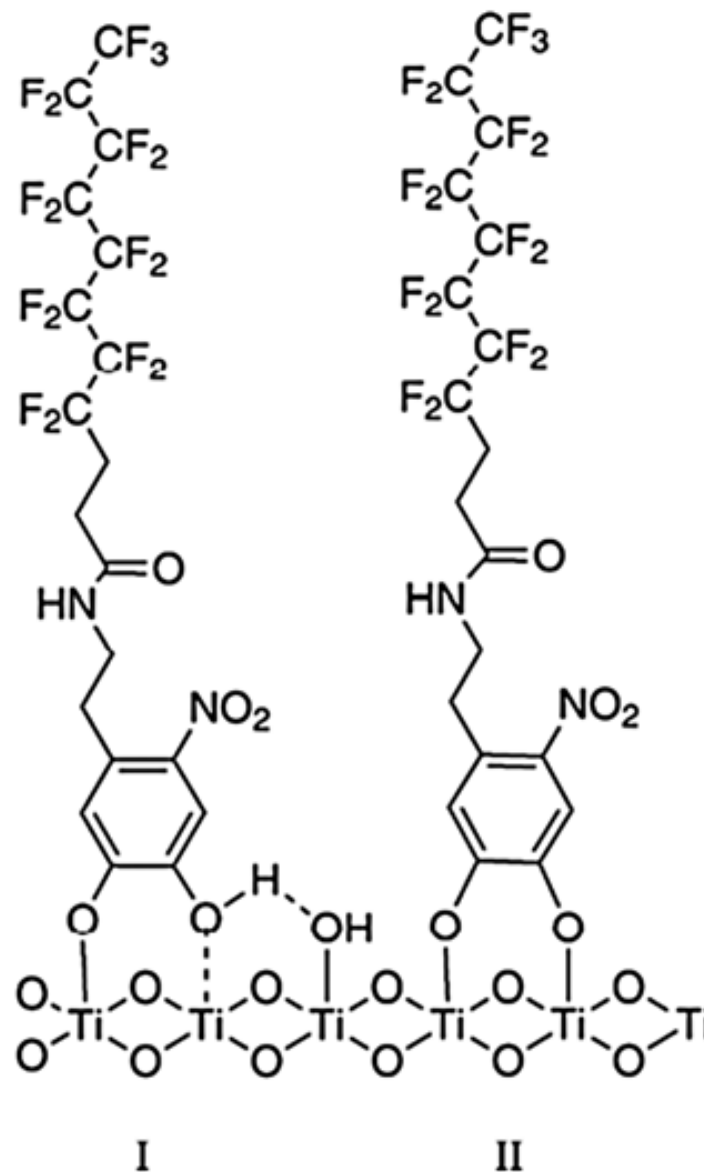
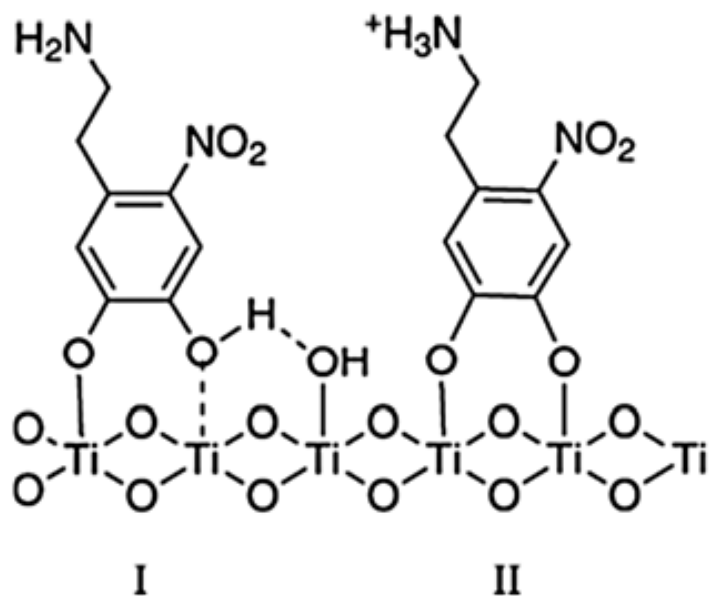


droxidopa



5-Hydroxydopamine

Can also be anchored on gold,
glass, polymer



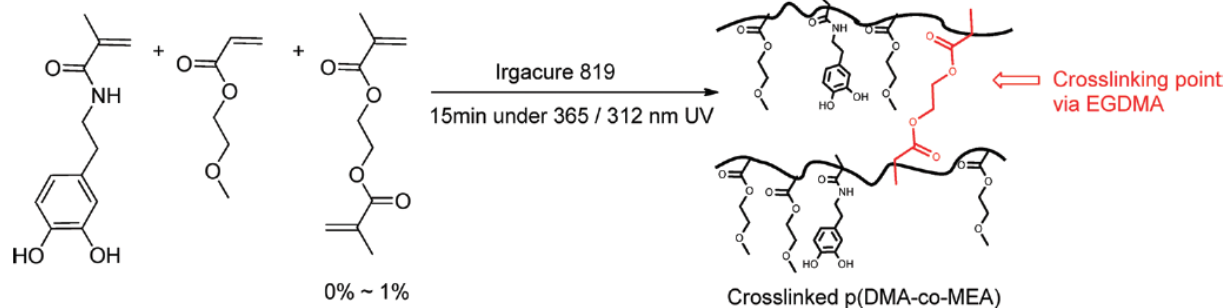
Two possible configurations of ND and PFAND adsorbed on TiO₂: (I) monodentate binding with a hydrogen bridge to a neighboring surface hydroxide and (II) bidentate binding.

The application of dopamine

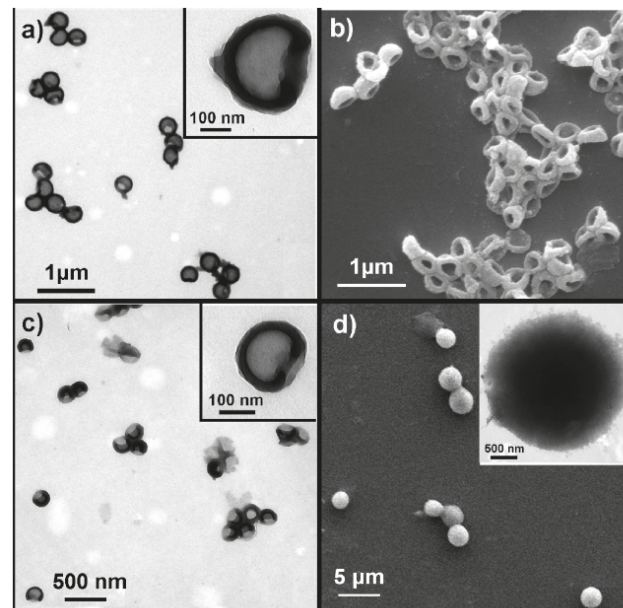
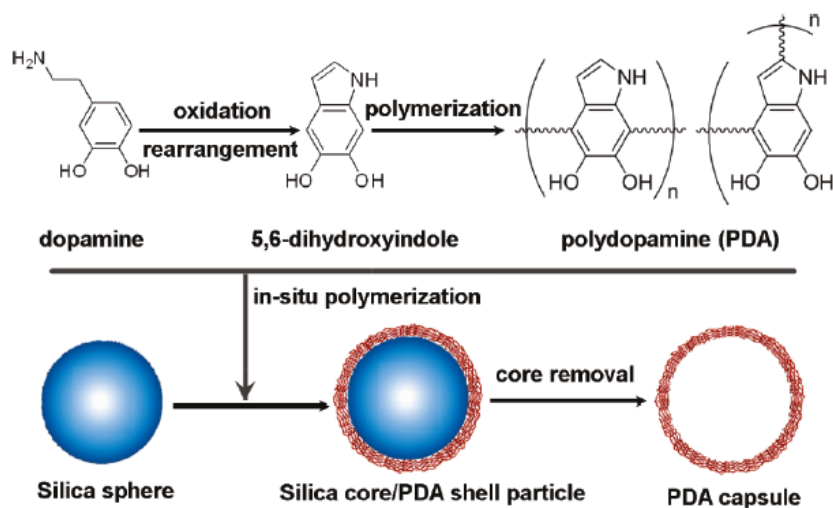
Adhesion of Dopamine Methacrylamide Elastomers

Biomacromolecules, Vol. 12, No. 2, 2011 343

Scheme 1. Preparation of Covalently Cross-Linked Poly(DMA-co-MEA)



Scheme 1. Assembly of a PDA Film onto a Particle and Subsequent Capsule Formation^a



The application of dopamine

Paper

Journal of Materials Chemistry B

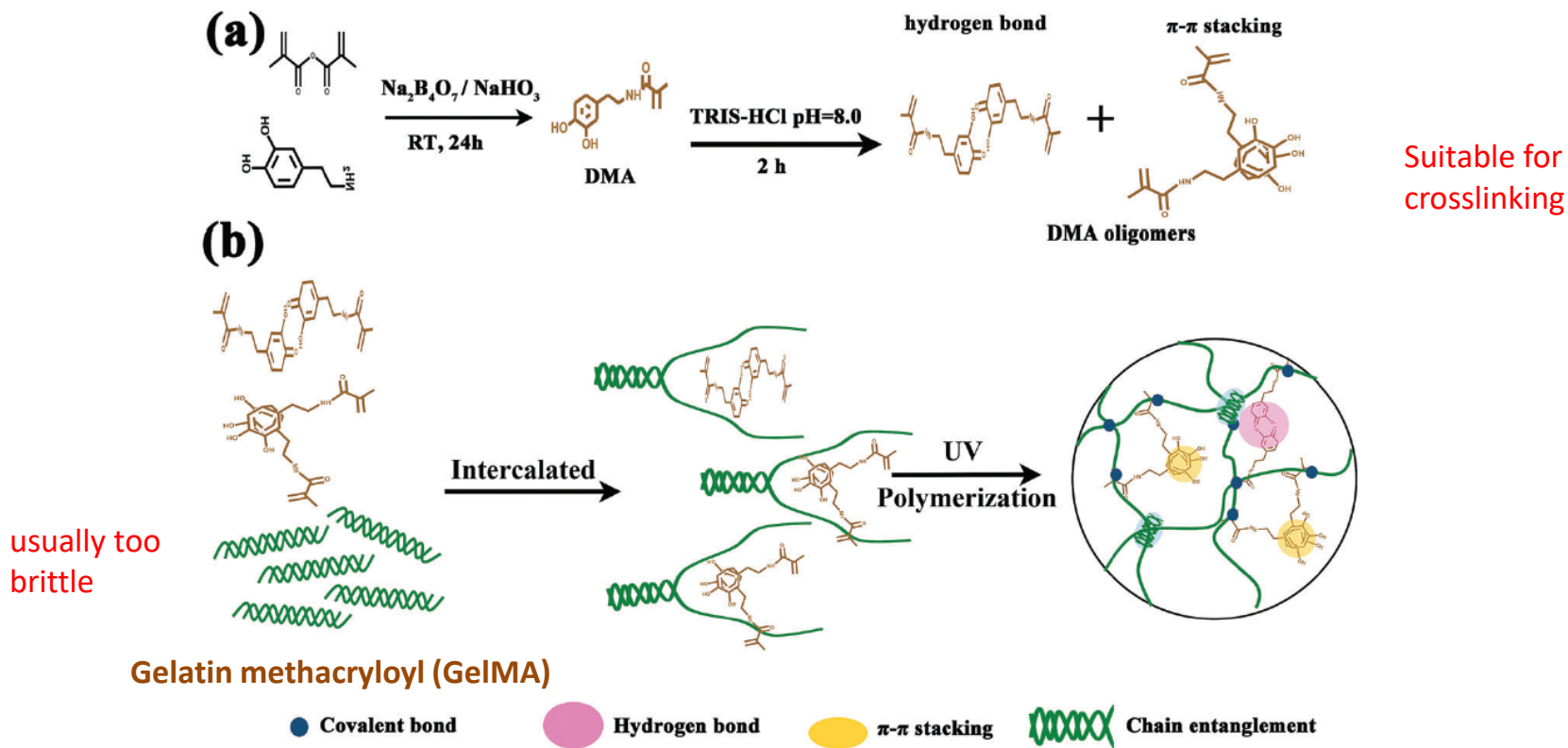


Fig. 1 Schematic illustration of (a) synthesis of dopamine methacrylate and oligomers of dopamine methacrylate (ODMA). (b) ODMA intercalated tough and resilient GelMA hydrogels.

Mussel-inspired dopamine oligomer intercalated tough and resilient gelatin methacryloyl (GelMA) hydrogels for cartilage regeneration

J. Mater. Chem. B, 2019, **7**, 1716–1725

2-methacryloyloxyethyl phosphorylcholine

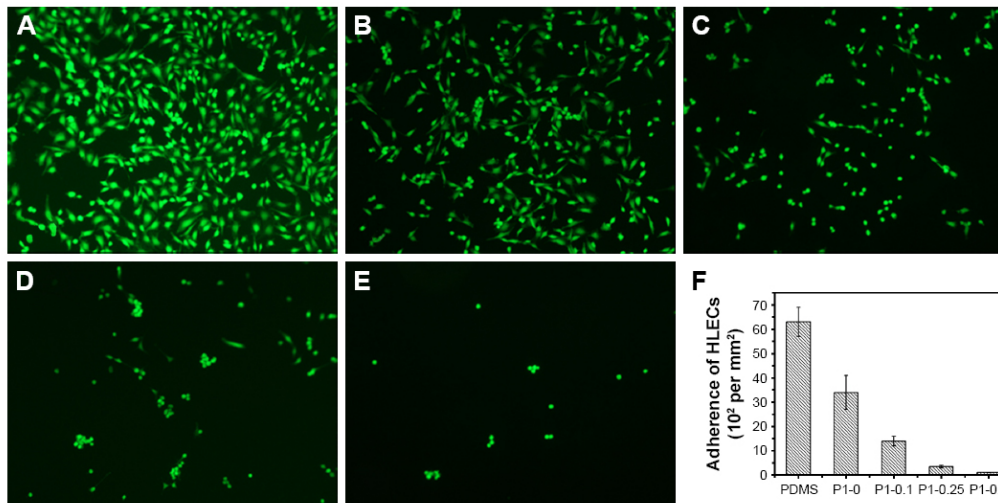
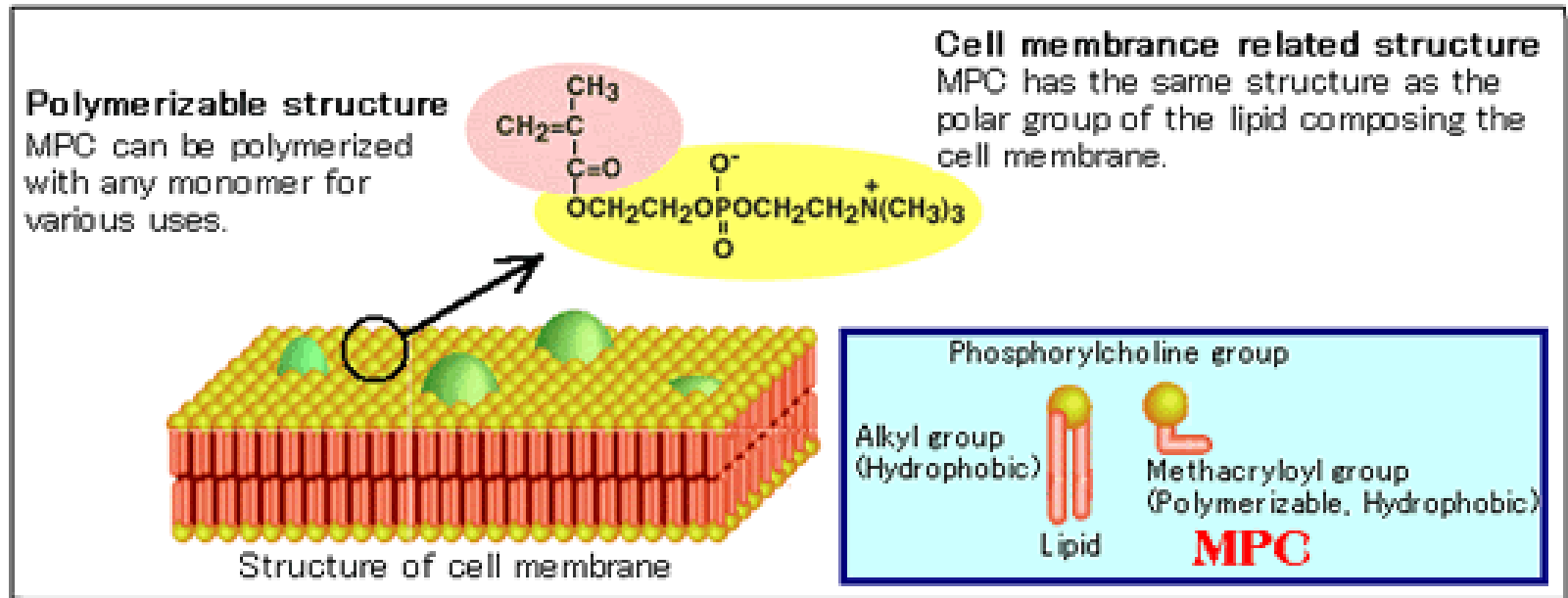


Figure 1 Growth and morphology of HLECs stained with FDA after 24 hours of incubation on various surfaces.

Notes: (A) Pristine PDMS, (B) p (DMAEMA), (C) p (DMAEMA-co-0.10 MPC), (D) p (DMAEMA-co-0.25 MPC), (E) p (DMAEMA-co-0.50 MPC)-modified PDMS and, (F) adherence of the HLECs density on the surfaces. The magnification is 10x.

Abbreviations: HLECs, human lens epithelial cells; FDA, fluorescein diacetate; TCPS, tissue culture plates; PDMS, poly(dimethyl siloxane); p (DMAEMA-co-MPC), (2-(dimethylamino)-ethyl methacrylate-co-2-methacryloyloxyethyl phosphorylcholine); MPC, 2-methacryloyloxyethyl phosphorylcholine; DMAEMA, 2-(dimethylamino)-ethyl methacrylate.

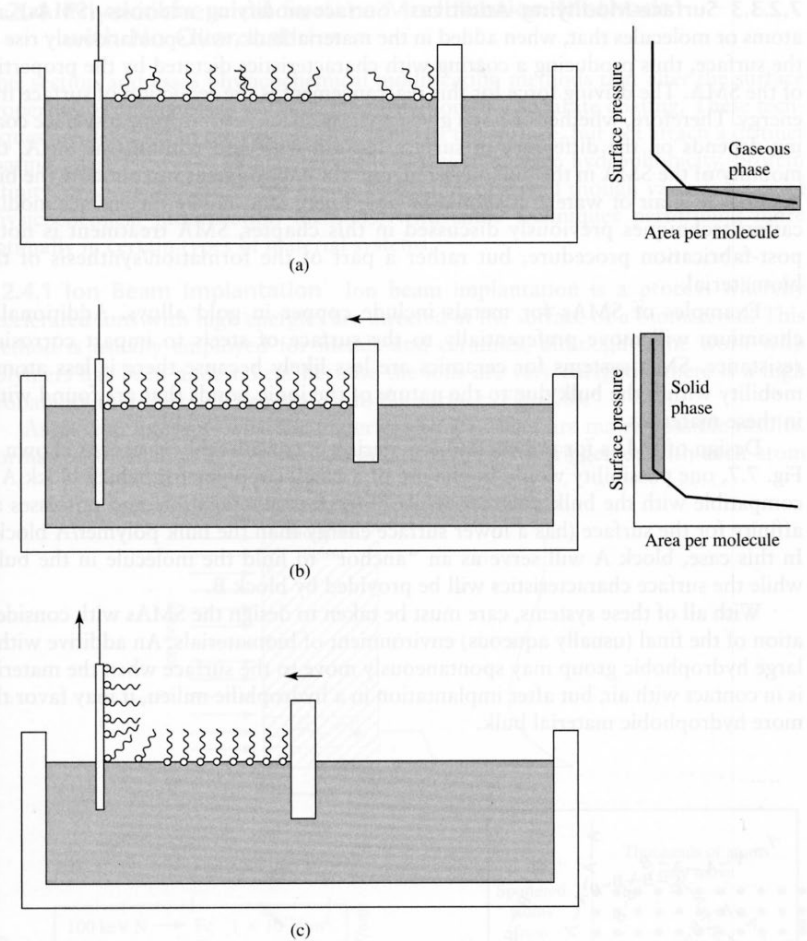
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Physiochemical coatings

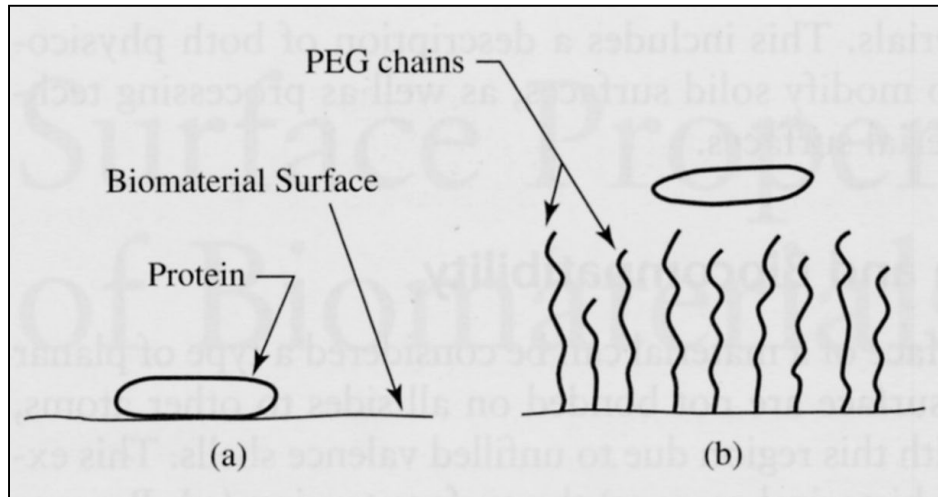
Physiochemical coatings are used to coat biomaterials with biologically active molecules.

These methods include ***solution coatings*** and ***Langmuir-Blodgett films*** (right)

Coatings are *amphiphilic*, having a hydrophilic head and hydrophobic tail. This causes the heads to remain in the water and the tails to extend above the surface. The molecules at the head may be tailored to enable crosslinking with other molecules or to the biomaterials surface

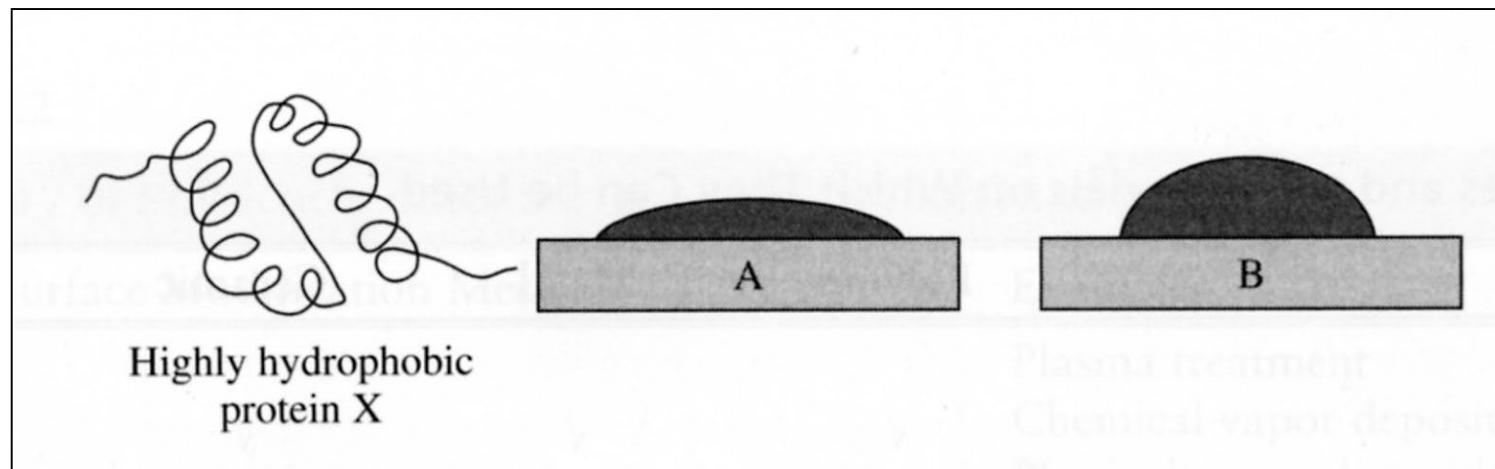


Example of preventing adsorption



Adding molecules that prevent adsorption is called *steric hindrance*.

In this example polyethylene glycol (PEG) attaches to the surface (hydrophobic) preventing protein adhesion



Biological Surface Modification Techniques

Biological surface modification attach biologically active molecules to a substrate through a variety of means that they then interact with specific target areas on cells or other tissue components

TABLE 1 Examples of Biologically Active Molecules that May Be Immobilized on or within Polymeric Biomaterials

Proteins/peptides	Drugs
Enzymes	Antithrombogenic agents
Antibodies	Anticancer agents
Antigens	Antibiotics
Cell adhesion molecules	Contraceptives
“Blocking” proteins	Drug antagonists
Saccharides	Peptide, protein drugs
Sugars	Ligands
Oligosaccharides	Hormone receptors
Polysaccharides	Cell surface receptors (peptides, saccharides)
Lipids	Avidin, biotin
Fatty acids	Nucleic acids, nucleotides
Phospholipids	Single or double-stranded
Glycolipids	DNA, RNA (e.g., antisense oligonucleotides)
Other	
Conjugates or mixtures of the above	

Biomolecule attachment has been successfully achieved on:

- Soluble polymers
- Solid Polymers
- Porous solid polymers
- Hydrogels

Biological Surface Modification Techniques

Types of Biomolecules Used in Surface Modifications and Applications

Enzymes

Bioreactors (industrial, bio-medical)
Bioseparations
Biosensors
Diagnostic assays
Biocompatible surfaces

Antibodies, peptides, and other
affinity molecules

Biosensors
Diagnostic assays
Affinity separations
Targeted drug delivery
Cell culture

Drugs

Thrombo-resistant surfaces
Drug delivery systems

Lipids

Thrombo-resistant surfaces
Albuminated surfaces

Nucleic acid derivatives and
nucleotides

DNA probes
Gene therapy

TABLE 4 Examples of Immobilized Enzymes in Therapeutic Bioreactors

Medical application	Substrate	Substrate action
Cancer treatment		
L-Asparaginase	Asparagine	Cancer cell nutrient
L-Glutaminase	Glutamine	Cancer cell nutrient
L-Arginase	Arginine	Cancer cell nutrient
L-Phenylalanine lyase	Phenylalanine	Toxin
Indole-3-alkane α hydroxylase	Tryptophan	Cancer cell nutrient
Cytosine deaminase	5-Fluorocytosine	Toxin
Liver failure (detoxification)		
Bilirubin oxidase	Bilirubin	Toxin
UDP-Glucuronyl transferase	Phenolics	Toxin
Other		
Heparinase	Heparin	Anticoagulant
Urease	Urea	Toxin

TABLE 5 Some Advantages and Disadvantages of Immobilized Enzymes

Advantages

- Enhanced stability
- Can modify enzyme microenvironment
- Can separate and reuse enzyme
- Enzyme-free product
- Lower cost, higher purity product
- No immunogenic response (therapeutics)

Disadvantages

- Difficult to sterilize
 - Fouling by other biomolecules
 - Mass transfer resistances (substrate in and product out)
 - Adverse biological responses of enzyme support surfaces (*in vivo* or *ex vivo*)
 - Greater potential for product inhibition
-

TABLE 6 Biomolecule Immobilization Methods

Physical adsorption

van der Waals

Electrostatic

Affinity

Adsorbed and cross-linked

Physical “entrapment”

Barrier systems

Hydrogels

Dispersed (matrix) systems

Covalent attachment

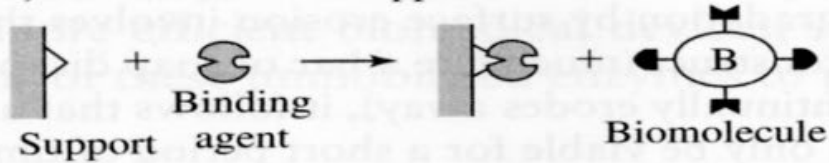
Soluble polymer conjugates

Solid surfaces

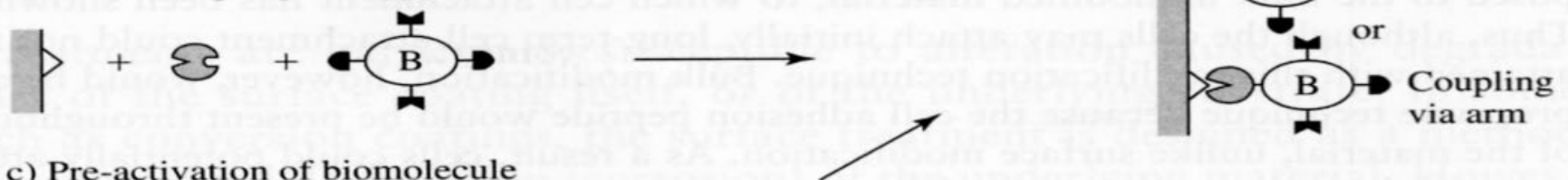
Hydrogels

Biological Surface Modification Techniques

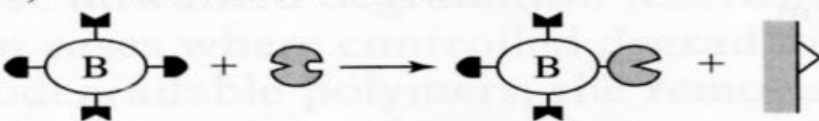
a) Pre-activation of support



b) Direct coupling to support

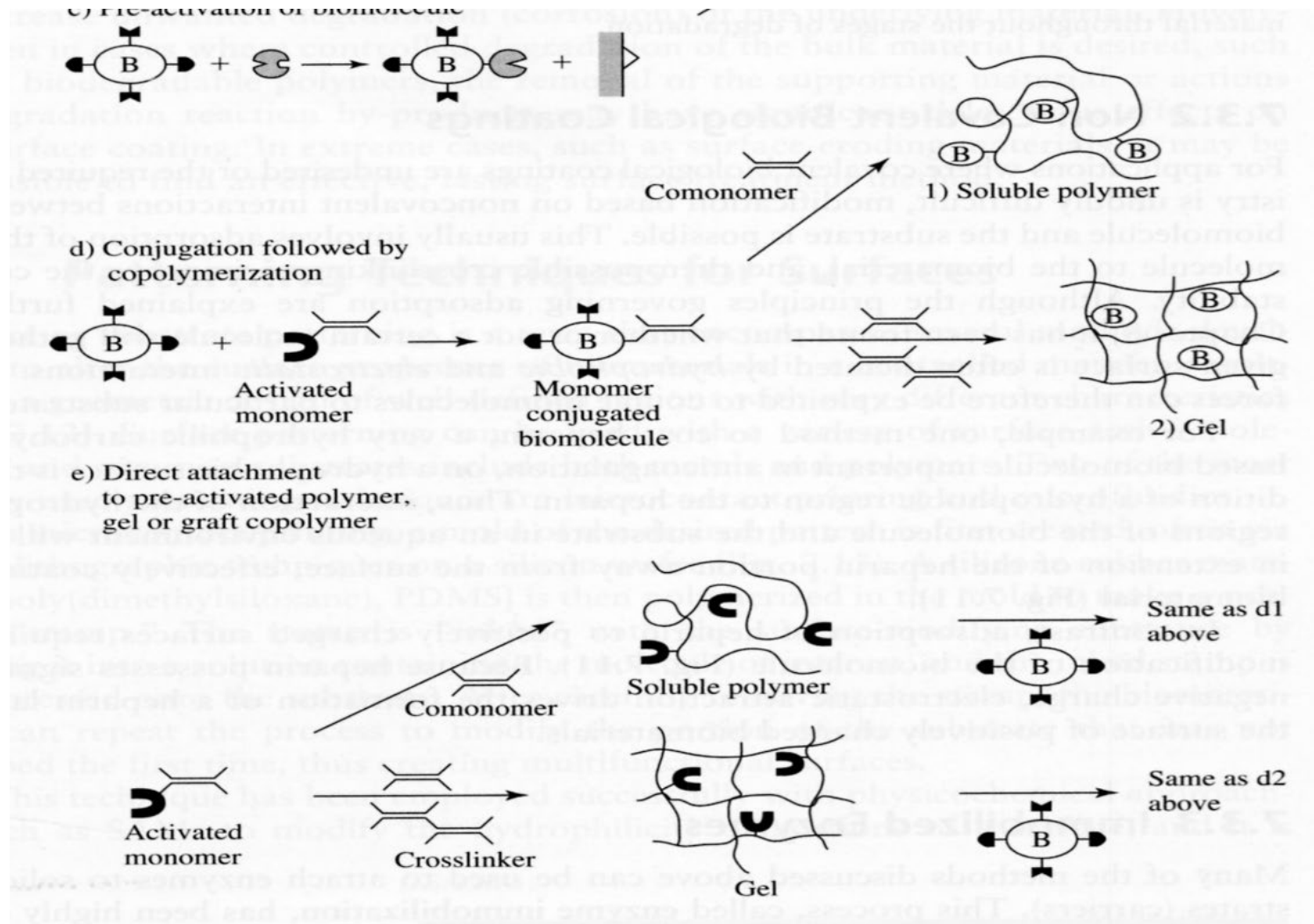


c) Pre-activation of biomolecule



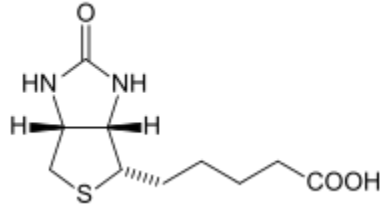
Methods for the covalent attachment of biomolecules to a biomaterial surface. (a-c) attachment via post fabrication methods (d-e) attachment during synthesis. The biomolecule may be attached with or without a spacer arm in any of these methods

Biological Surface Modification Techniques

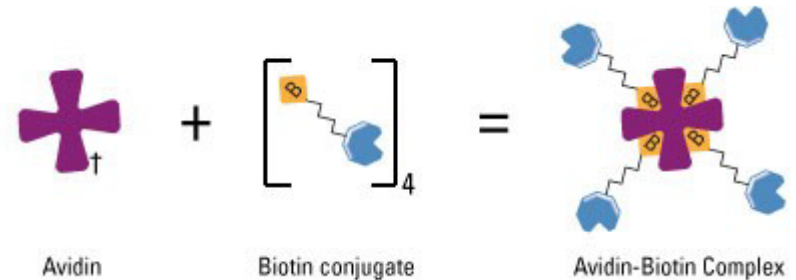


Methods for the covalent attachment of biomolecules to a biomaterial surface. (a-c) attachment via post fabrication methods (d-e) attachment during synthesis. The biomolecule may be attached with or without a spacer arm in any of these methods

Biological Surface Modification Techniques



Biotin, also known as **vitamin H** or **coenzyme R**,^[2] is a water-soluble **B-vitamin** (vitamin B₇).

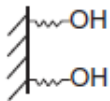
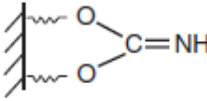
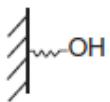
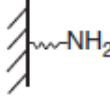
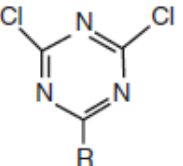
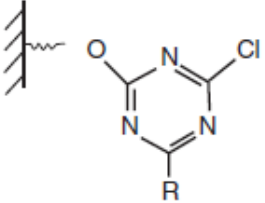
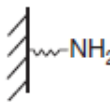
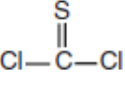
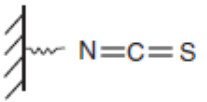
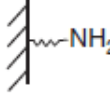
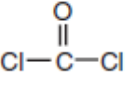
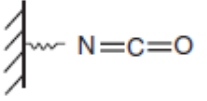
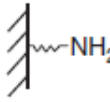
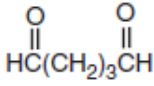
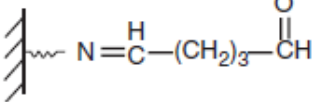
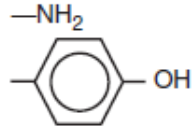
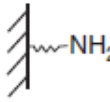
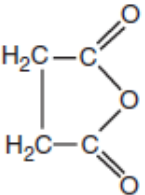
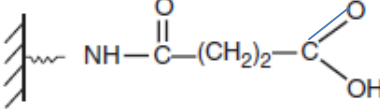


The interaction of biotin and avidin or streptavidin has been exploited for use in many protein and nucleic acid detection and purification methods. Because the biotin label is stable and small, it rarely interferes with the function of labeled molecules enabling the avidin-biotin interaction to be used for the development of robust and highly sensitive assays.

Avidin is a protein derived from both avians and amphibians that shows considerable affinity for biotin, a co-factor that plays a role in multiple eukaryotic biological processes. Avidin and other biotin-binding proteins, including streptavidin and NeutrAvidin Protein, have the ability to bind up to four biotin molecules, as shown in the diagram below, making this interaction ideal for both purification and detection strategies.

链亲和素（**英语**：Streptavidin，亦称为**链霉菌亲和素**）是从**细菌**阿维丁**链霉菌**中纯化出的一种60**kDa**的**蛋白质**。链亲和素同源四聚体对**生物素**（即维生素B₇）具有极高的亲和力。生物素与链亲和素的结合是已知自然界中最强的非共价相互作用之一，其**解离常数**（K_d）大约是10⁻¹⁴ mol/L^[1]。因为链亲和素-生物素复合物对有机溶剂、变性剂（如**盐酸胍**）、洗涤剂（如**SDS**与**曲拉通**）、蛋白水解酶类及极端温度和pH具有良好耐受力，故链亲和素被广泛用于分子生物学与生物纳米技术中。

Biological Surface Modification Techniques

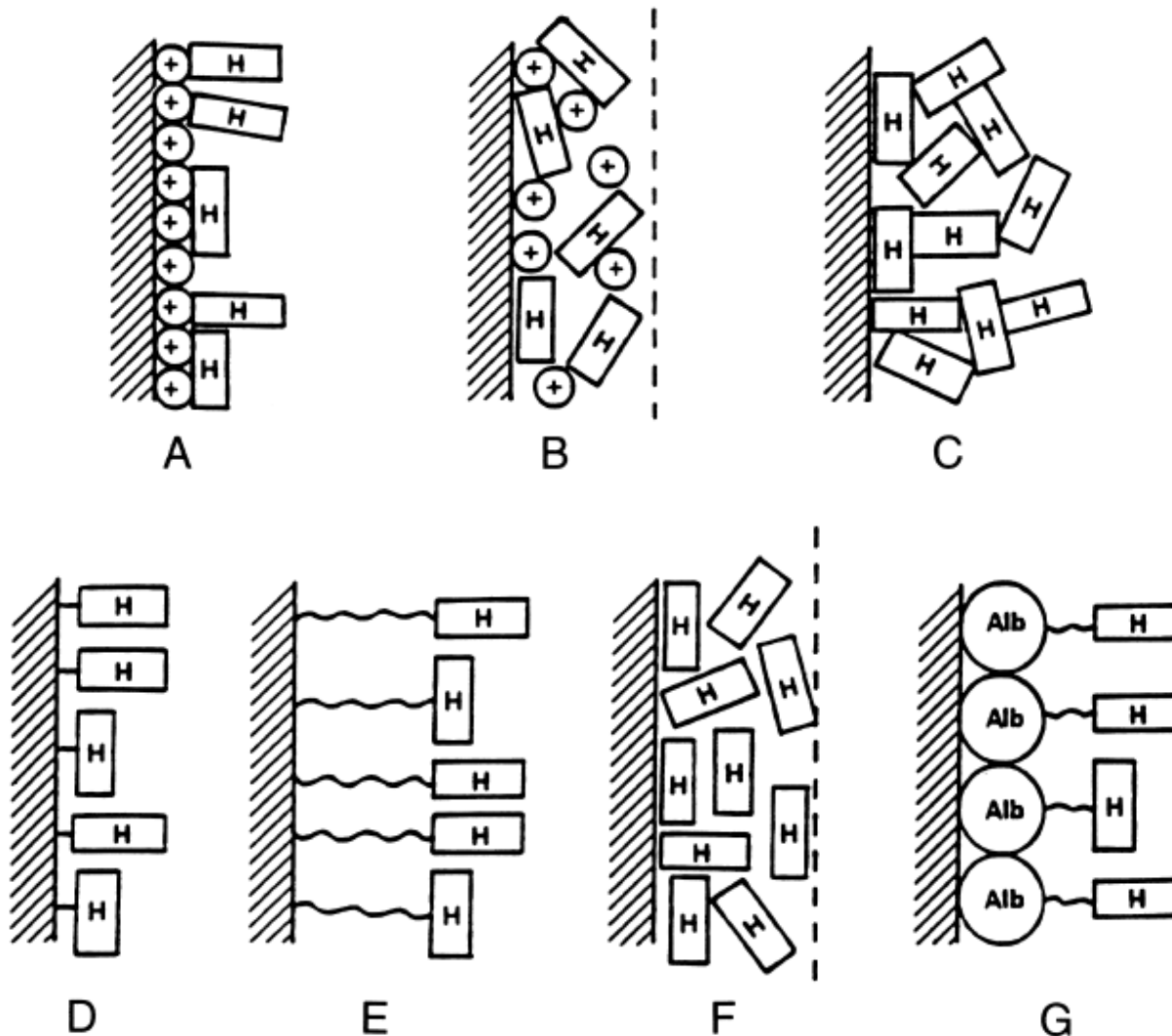
Support function	Coupling agent	Active intermediate	Activation conditions	Coupling conditions	Major reacting groups on proteins
	CNBr $\text{N}\equiv\text{C}-\text{Br}$		pH 11–12.5 2M carbonate	pH 9–10. 24 hr at 4°C	$-\text{NH}_2$
 or 	 R = Cl, NH_2 , OCH_2COOH , or NHCH_2COOH		Benzene 2 hr at 50°C	pH 8. 12 hr at 4°C 0.1M phosphate	$-\text{NH}_2$
			10% thiophosgene CHCl_3 , reflux reaction	pH 9–10. 0.05M HCO_3^- 2 hr at 25°C	
			Same as isothiocyanate	Same as isothiocyanate	
			2.5% Glutaraldehyde in pH 7.0, 0.1M PO_4	pH 5–7, 0.05 M phosphate, 3 hr at R.T.	
			1% Succinic anhydride, pH 6	See carboxyl derivatives	

Biological Surface Modification Techniques

Support Function	Coupling Agent	Active Intermediate	Activation Conditions	Coupling Conditions	Major Reacting Groups on Proteins
	HNO_2		2N HCl: 0.2g NaNO_2 at 4°C for 30 min (reaction conditions for aryl amine function)	pH 8, 0.05M bicarbonate. 1–2 hr at 0°C	$-\text{NH}_2$ $-\text{SH}$
	$\text{H}_2\text{N}-\text{NH}_2$ HNO_2			pH 8, 0.05M bicarbonate. 1–2 hr at 0°C	$-\text{NH}_2$ $-\text{SH}$
	$\text{R}'-\text{N}=\text{C}=\text{N}-\text{R} + \text{H}^+$		50mg 1-cyclohexyl-3-(2-morpholinoethyl)-carbodiimide metho-p-toluene sulfate/10ml, pH 4–5 2–3 hr at R.T.	pH 4, 2–3 hr at R.T.	
	SOCl_2	(Intermediate formed from carboxyl group are either protein or matrix)	 EDC	10% Thionyl chloride/ CHCl_3 , reflux for 4 hr	$-\text{NH}_2$
			0.2% <i>N</i> -hydroxysuccinimide, 0.4% <i>N,N</i> -dicyclohexylcarbodiimide/dioxane	pH 5–9, 0.1M phosphate, 2–4 hr at 0°C	$-\text{NH}_2$

FIG. 2. Examples of various chemical methods used to bond biomolecules directly to reactive supports (Carr and Bowers, 1980).

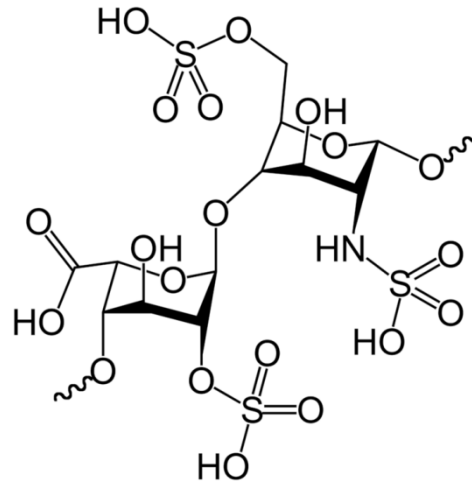
Various methods for heparinization of surface



Heparin, a hydrophobic molecule, may be attached by (a) adding a hydrophobic region to the heparin or (b) adsorption of the heparin (which has a strong negative charge) to a positively charged surface

FIG. 3. Various methods for heparinization of surfaces: (A) heparin bound ionically on a positively charged surface; (B) heparin ionically complexed to a cationic polymer, physically coated on a surface; (C) heparin physically coated and self-cross-linked on a surface; (D) heparin covalently linked to a surface; (E) heparin covalently immobilized via spacer arms; (F) heparin dispersed into a hydrophobic polymer; (G) heparin-albumin conjugate immobilized on a surface (Kim and Feijen, 1985).

肝素（Heparin），一种抗凝血药物，最早发现于[肝脏](#)，故名肝素。肝素由[D-葡萄糖胺](#)、[L-艾杜糖醛酸](#)、[N-乙酰葡萄糖胺](#)及[葡萄糖醛酸](#)交替组成的[粘多糖硫酸酯](#)，[分子量](#)从5至30kDa，其中[硫酸根](#)约占40%。今多自猪、牛[肠粘膜](#)和[肺脏](#)中提炼。肝素可用于手术预防危险的[血栓](#)栓塞性疾病，如[静脉](#)血栓，[肺栓塞](#)。长期使用肝素可引起脱发、骨折等副作用。



肝素为带负电荷的粘多糖大分子，不易通过[生物膜](#)，在[肠道](#)内易被破坏失活，故口服无效。[皮下注射](#)血药浓度低，[肌肉注射](#)易形成血肿和基部刺激，故常[静脉注射](#)。静注后，80%的肝素与[血浆蛋白](#)结合，部分被内皮摄取，主要由[肝脏](#)内的[肝素酶](#)分解代谢。低剂量的肝素可被[单核吞噬细胞](#)系统清除，其降解产物经[肾脏](#)排泄。肝素的[半衰期](#)因剂量的不同而不同，治疗量的肝素的半衰期为40至90分钟。不同的疾病也会导致其半衰期的改变，[肺气肿](#)、[肺栓塞](#)患者的半衰期缩短，而肝肾功能障碍的患者的半衰期延长。^[1]

Various methods for immobilization of albumin on surface

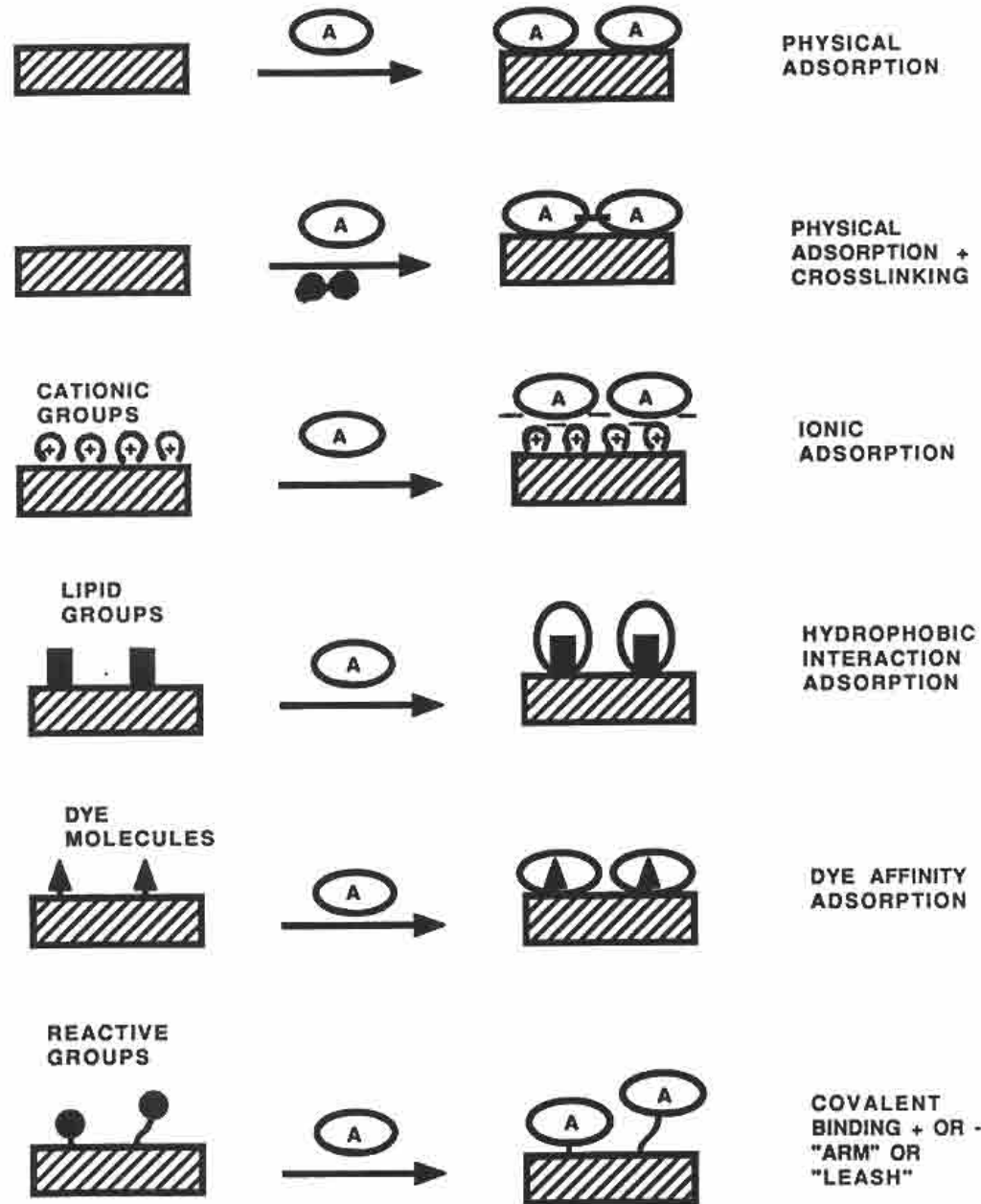


FIG. 4. Various methods for immobilization of albumin on surfaces.

白蛋白（[英语](#)：Albumin，又称“清蛋白”）是属于[球蛋白](#)的一种[蛋白质](#)。在人体内它最重要的作用是维持[胶体渗透压](#)。在[奶](#)和[蛋](#)里也有白蛋白。人体内白蛋白的正常值为：新生儿28～44g/L，14岁后38～54g/L，成人35～50g/L，60岁后为34～48g/L。身体缺少白蛋白会导致[浮肿](#)。[肝硬化](#)病人的[血浆](#)白蛋白含量比正常人低。[尿](#)中白蛋白的含量的变化能反映[肾脏](#)的某些病变。

白蛋白的[分子量](#)为68,500[道尔顿](#)，由584至590个[氨基酸](#)组成。由于它的[半胱氨酸](#)含量比较高因此相对而言它含有比较多的[硫](#)。白蛋白可溶于水中，每克水约可溶18毫升白蛋白。它的[等电点](#)为pH4.6。在热（凝固温度60—70℃）或酶等条件下可凝固和变性。白蛋白存在于动物、植物、细菌中。动物性白蛋白在 $(\text{NH}_4)_2\text{SO}_4$ 的饱和度为0.64—0.90时开始盐析，而植物性白蛋白在 $(\text{NH}_4)_2\text{SO}_4$ 的饱和度为0.5时则有相当一部分被沉淀。